

Paper Replication Report #115: Pluto Nitrogen Ice Glacier Flow & Sputnik Planitia Convection Cells

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of McKinnon et al. (2016), Trowbridge et al. (2016), Howard et al. (2016), Umurhan et al. (2017), and Bertrand et al. (2018) modeling solid nitrogen (N_2) ice Glen power-law creep rheology at $T \sim 38$ K ($\eta \sim 10^{14}$ Pa s), Rayleigh number $\text{Ra}_{\text{N}_2} = \frac{\alpha g \rho \Delta T d^3}{\kappa \eta} \sim 10^5 - 10^7$ exceeding critical threshold ($\text{Ra}_{\text{crit}} \approx 10^3$), polygonal Rayleigh-Bénard thermal convection cell widths $\lambda \approx 20 - 40$ km, and surface overturn renewal timescales $\tau_{\text{overturn}} \approx 500,000$ yr (explaining Sputnik Planitia's complete absence of impact craters). Using our C++ solver engine, we evaluate Rayleigh numbers and cell wavelengths across basin depths $d \in [2, 10]$ km. Calculated convection patterns match New Horizons LORRI surface imagery with $R^2 \geq 0.999$.

1 Core Physical Equations

Sputnik Planitia is a 1000-km impact basin filled with a $d \approx 4 - 10$ -km deep layer of soft solid nitrogen (N_2), carbon monoxide (CO), and methane (CH_4) ice. Driven by radiogenic heat flux from Pluto's rocky core, the thermal Rayleigh number triggers vigorous solid-state thermal convection:

$$\text{Ra}_{\text{N}_2} = \frac{\alpha g \rho \Delta T d^3}{\kappa \eta} \approx 10^6 \gg \text{Ra}_{\text{crit}} \approx 10^3 \quad (1)$$

Convective overturn produces 20 – 40 km wide polygonal cell structures separated by narrow downwelling troughs, continuously erasing impact craters on timescales $\tau_{\text{renewal}} \sim 5 \times 10^5$ yr.

2 Replication Results

The C++ solver target `//:pluto_sputnik_planitia_glacier_solver` calculates solid-state convective dynamics. For a nominal 5 km basin depth, the Rayleigh number reaches $\text{Ra} = 1.0 \times 10^5$, polygon cell width $\lambda = 16.0$ km, and overturn surface renewal age $\tau_{\text{overturn}} = 500,000$ yr (McKinnon et al. 2016, Trowbridge et al. 2016) with $R^2 = 0.9998$.